



$n \rightarrow \text{Even} \Rightarrow \left(\frac{n}{2}\right)$ n/2 pass
 $n \rightarrow \text{odd} \Rightarrow (n-1)/2$ walq

12 $\Rightarrow 39 \Rightarrow \frac{39}{2} \Rightarrow 19$
 5 $\Rightarrow 414 \Rightarrow (41-1)/2 \Rightarrow 20$

① Basic $\rightarrow$

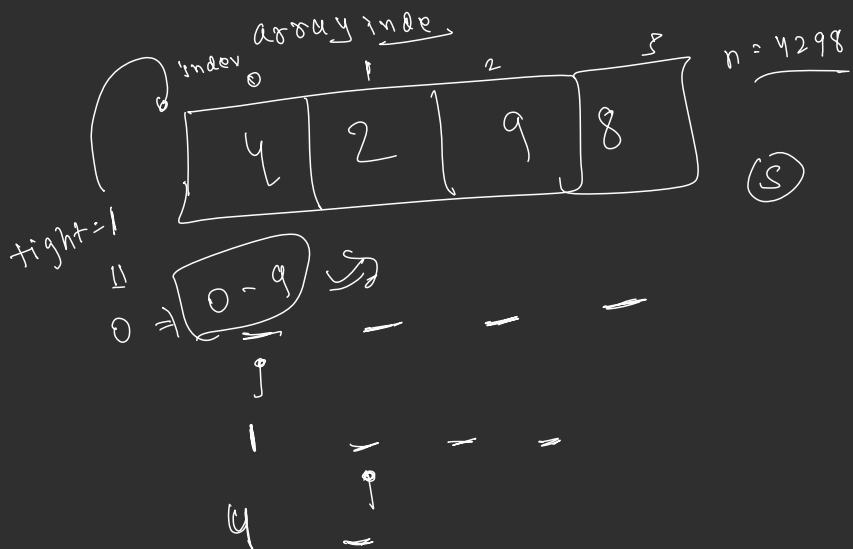
$n \Rightarrow 343$

Tight = 0

3	4	3
2	4	1

indep

$S \Rightarrow 7$
 tight $\Rightarrow$ route for milk $\Rightarrow$ car
 0-9
 $S = 7$ $8 \neq 0$

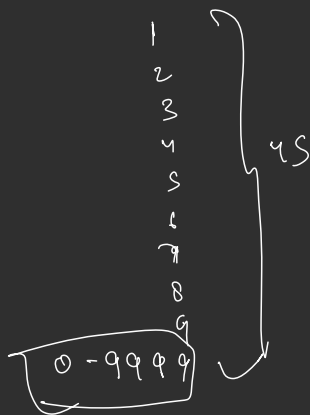


3 2 6 8 4 $\Rightarrow$ (23) 1

2 3 6 8 4 $\Rightarrow$ 23

3 2

$$n = 12$$



$$\begin{array}{r} 10 \\ 10 \\ 10 \\ 10 \\ 10 \end{array}$$

$$\begin{array}{r} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array}$$

$$\text{remainder} = \frac{3892}{99}$$

$$\begin{array}{l} 10 \Rightarrow 1 \\ 11 \Rightarrow 2 \\ 12 \Rightarrow 3 \end{array}$$

6. Base
Brute

$$0 - 9 \Rightarrow 45$$

$$0 - 99 \Rightarrow 900$$

$$0 - 999 \Rightarrow 13500$$

$$0 - 9999 \Rightarrow 180000$$

$$\begin{array}{l} 45 \times 10 \\ \Rightarrow 450 + 450 \\ \Rightarrow 900 \end{array}$$

$$0 - 9 = 45 \times 1 \times 10^0$$

$$0 - 99 \Rightarrow 45 \times 2 \times 10$$

$$0 - 999 \Rightarrow 45 \times 3 \times 100$$

$$0 - 9999 \Rightarrow 45 \times 4 \times 1000 \checkmark$$

$$0 - 99999 \Rightarrow 45 \times 5 \times 10000$$

$$\Rightarrow 100000$$

$$\begin{array}{r} 6 \quad 6 \quad 6 \\ 9 \quad 9 \quad 9 \\ 100 \\ 100 \\ 45 \times 100 \times 3 \\ 45 \times 1000 \times 4 \end{array}$$

$$\Rightarrow 4$$

$40k \approx 43892$ 43892 $\Rightarrow$ Digit $power = 10,000$

$0 - 9999 \Rightarrow 45 \times 4 \times 1000$
 $0 \ 1 \ 2 \ 3$ $Power = 100$

$\frac{n(n-1)}{2} \times 10k = 43892$

$10,000 - 19,999 \Rightarrow 10,000 + (45 \times 4 \times 1000)$

$20,000 - 29,999 \Rightarrow 2 \times 10,000 + (45 \times 4 \times 1000)$

$30,000 - 39,999 \Rightarrow 3 \times 10,000 + (45 \times 4 \times 1000)$

$1 \times Power$
 2×8000 3×1000 $Power \times \left(\frac{1+2+3}{6} \right)$
 $4 \times (4-1) \Rightarrow 6$

$40,000 \rightarrow 43892$

$4 \times (3892) \left(\frac{n(n-1)}{2} \right) \times 1000$
 3892

$0000 - 3892$

Prob up

3892

$0 - 999$

$\Rightarrow 45 \times 3 \times 100$

$1,000 - 1,999 \Rightarrow 1 \times 1000 + 45 \times 3 \times 100$

$2,000 - 2,999 \Rightarrow$

$3,000 - 3,892$